Predictive Analytics for Optimizing Sanitaryware Retail : Ensuring Product Completeness and Pricing

A Final report for the BDM capstone Project

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Executive Summary

The primary objective of this project was to analyze the operational and financial challenges of "K*** A****** Store" ,focusing specifically on three critical business pillars: Credit Management, Pricing Fluctuations (MRP Changes), and Inventory Losses. The business operations are heavily influenced by credit-based transactions, where a small group of high-priority customers holds a significant share of the shop's outstanding payments.

Delays in updating selling prices to revised MRPs reduced profit margins. Inventory losses are concentrated in fragile premium sanitaryware items, driving disproportionate financial losses.

The study further highlighted that business performance peaked in earlier years and has since shown fluctuations, emphasizing the need for continuous monitoring and adaptive strategies. Pricing analysis indicated an overall increase in inventory valuation, driven largely by adjustments in premium product segments.

Furthermore, the analysis demonstrated the importance of applying the Pareto principle in business decision-making, where a small proportion of customers and products account for the majority of impact. By identifying these “vital few” elements, the business can prioritize efforts more effectively, improve resource allocation, and focus on areas that yield the highest returns in terms of revenue optimization and loss reduction.

Based on these insights, several actionable recommendations were proposed, including prioritizing debt recovery from high-risk customers, implementing stricter credit control policies, aligning selling prices promptly with MRP changes, improving handling and storage of fragile goods, and focusing on high -value product categories for better inventory management.

Overall, this project demonstrates how data-driven decision-making can significantly enhance operational efficiency, reduce financial risks, and support sustainable business growth in a retail environment

Link to dataset - Click here

Detailed Explanation of Analysis Process/Method

❖ Introduction

- The analysis of the collected data is carried out using a structured analytical approach that combines both quantitative and qualitative techniques. The datasets used in this project include credit records , MRP changes and broken parts .

- To systematically analyze the data, the following four types of analytical methods are adopted:

  ➔ Descriptive analysis
  ➔ Diagnostic analysis
  ➔ Predictive analysis
  ➔ Prescriptive analysis

- Descriptive analysis summarizes past data to show what has happened in the business. Key insights like outstanding payment distribution and pricing differences were earlier presented in the 'Results and Findings' section of the mid-term report.

- Diagnostic analysis identifies reasons behind observed patterns. Using spreadsheets, sorting, and Power BI visuals, it explains issues like debt concentration, pricing inconsistencies, and product losses, discussed further in this report.

- Predictive analysis forecasts future trends using past historical data patterns.

- Prescriptive analysis suggests practical actions. Based on insights, steps like focused debt recovery, pricing revisions, and better controls are recommended to enhance business performance.

❖ Objective

- The primary objective of this project is to analyze operational and financial challenges of the retail store, particularly focusing on credit management, pricing fluctuations, and product losses.

- In addition to working with raw transactional data, an effort has been made to understand customer behavior, payment patterns, and operational inefficiencies to provide practical and implementable solutions.

- Through detailed discussions with the shop owner, it was identified that a significant portion of the business operations is influenced by credit customers. A small group of customers contributes to a large share of outstanding payments, making them highly important for cash flow management.

- The credit records maintained in the traditional register were analyzed to track borrowing details, partial payments, and remaining balances. This helped in understanding transaction patterns and how payments typically remain unsettled.

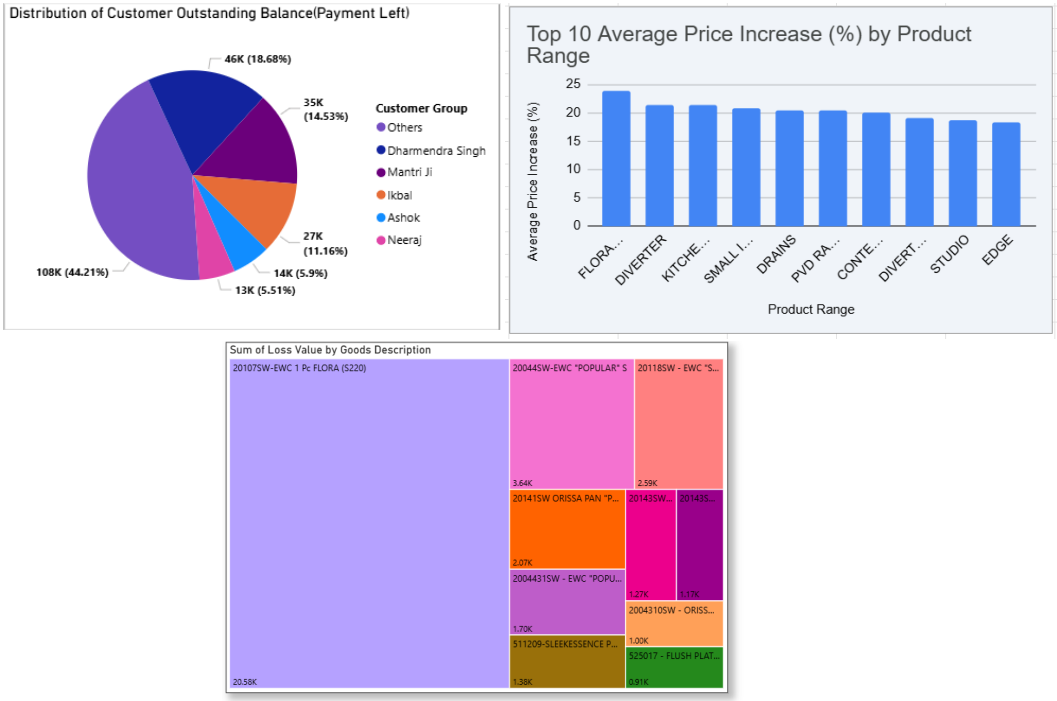


Figure 1

- The focus of this analysis was therefore to understand payment behavior patterns, identify high-risk customers, and determine how these patterns affect the overall financial stability of the shop. Special attention was given to customers with large pending balances, as delays from these accounts have the highest impact on working capital.

- These datasets were collectively used to create a structured understanding of the business workflow. This helped in forming a clear understanding of credit cycles, pricing updates, and product handling processes, which further supported better decision-making and operational planning.

❖ Pre-Processing of the Data

- Collecting the data for this project was a challenging task as the records were not maintained in a fully structured digital format. Most of the information was available in physical registers, supplier documents, and through direct discussions with the shop owner. These sources were carefully converted into sheets for further analysis.

- The datasets primarily included details such as customer names, transaction dates, payment amounts, outstanding balances, product descriptions, MRP values, and broken quantity information. Since the data was fragmented across multiple sources, it required proper organization before any meaningful analysis could be performed.

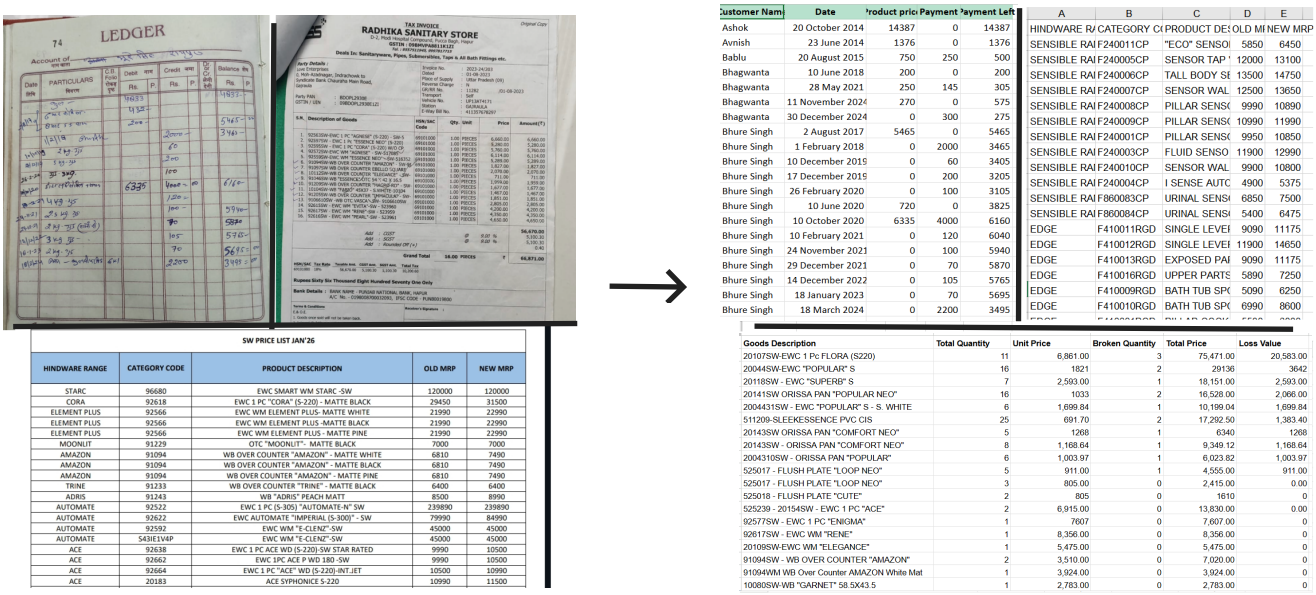


Figure 2

- For pricing data, the old and new MRP values were aligned in a structured format to enable easy comparison. This made it possible to track price changes across different product categories and identify significant variations.

  In the case of broken parts data, calculated fields such as total price and loss value were created using simple multiplication formulas. This helped in quantifying the financial impact of damaged or missing items.

  Overall, the preprocessing stage ensured that raw and unstructured data was transformed into a clean, consistent, and analysis-ready format. This step was essential for enabling accurate calculations, meaningful visualizations, and reliable insights in the later stages of the project.

- The analysis was carried out on a customer-wise basis. Transaction data was originally recorded on a periodic basis, including purchase amounts and payments made. This data was cleaned and organized, then aggregated to calculate total purchase amount, total payments made, and the remaining balance for each customer by using formulas and pivot tables.

- Pivot tables were further used to summarize the transaction data on a customer-wise basis. Since the data included multiple entries for purchases and payments on different dates, all records for each customer were grouped together to calculate the total purchase amount, total payments made, and the remaining balance. Any discrepancies or split entries were carefully reviewed and adjusted, and finally, a consolidated summary table for each customer was prepared. The Snapshot is shown below , can view the whole table by Click on here.

Customer Name	Total Product price	Total Payment	Total Payment Left
Dharmendra Singh	247182	201613	45569
Mantri Ji	35444	0	35444
Ikbal	67215	40000	27215
Ashok	14387	0	14387
Neeraj	13446	0	13446
Yogendra Singh	21916	13330	8586
Deshraj Singh	9044	500	8544
Samerpal Singh	18241	10000	8241
Shavir	8140	0	8140
Vikki Singh	24560	16800	7760
Deva	21035	15035	6000
Kashi	6856	1000	5856
Kartara	16330	10700	5630
Harveer Singh	5505	0	5505
Kamal singh	9981	5500	4481
Gopal Singh	4310	0	4310
Jitendra Singh	6078	2500	3578
Madan Singh	5497	2000	3497
Bhure Singh	12520	9025	3495
Narendra singh	3249	0	3249
Rajeev	3880	1100	2780
Jagesh	4464	2000	2464
Chopra	3380	1565	1815
kartar singh	1953	250	1703

Figure 3

❖ Descriptive Analysis

- With the help of spreadsheet tools, various formulas were used to generate summary statistics from the datasets, which played a crucial role in understanding the overall data structure and key patterns within it.

- To make the data more understandable and interpretable, visual representations were created using Power BI. These visualizations simplified complex datasets and allowed for quicker identification of important insights.

- To analyze the distribution of outstanding balances among customers, pie charts were used to highlight the proportion of total debt held by different customer groups. This helped in identifying the most critical accounts contributing to the majority of pending payments.
- Similarly, visual comparisons were created to examine the contribution of different product categories in terms of pricing changes and financial impact. These comparisons helped in identifying high-impact products and categories.
- Column charts were used to compare values across different categories, such as product ranges and customer groups, making it easier to observe differences and highlight key areas of concern.
- Descriptive statistics such as maximum values, minimum values, and total sums were calculated using spreadsheet formulas. These calculations were applied across datasets like credit records, MRP changes, and broken parts data to provide a clear numerical summary of the analysis.

❖ Diagnostic Analysis

- Diagnostic analysis focused on identifying root causes of key business issues such as delayed customer payments, price inconsistencies, and losses from damaged or missing products. Trends in datasets, combined with discussions with the shop owner, helped uncover patterns behind these problems.
- Analysis of credit records revealed that a significant portion of outstanding payments was concentrated among a few customers, leading to cash flow challenges. Pricing data showed that delays in updating selling prices (based on revised MRP) caused products to be sold at outdated rates, reducing profit margins.
- Examination of the broken parts dataset highlighted operational inefficiencies, where a small number of products contributed disproportionately to total losses due to handling and transportation issues. Visualizations and summary tables supported these insights, ensuring a clear understanding of these problems and their underlying causes.

Results and Findings

The insights drawn from the analysis of the data are based on three problem statements ,lets begin with

"DELAY IN CUSTOMER PAYMENTS"

➢ To identify the customers contributing the highest share in the total outstanding balance, sorting and filtering techniques were applied to the credit dataset, resulting in a structured summary table.

Customer Group	Payment Left
Others	107834
Dharmendra Singh	45569
Mantri Ji	35444
Ikbal	27215
Ashok	14387
Neeraj	13446
Total	243895

Figure 4

- Visual representation of this data provides clearer insights into the distribution of outstanding balances among customers. Therefore, the following chart was generated from the summarized data.

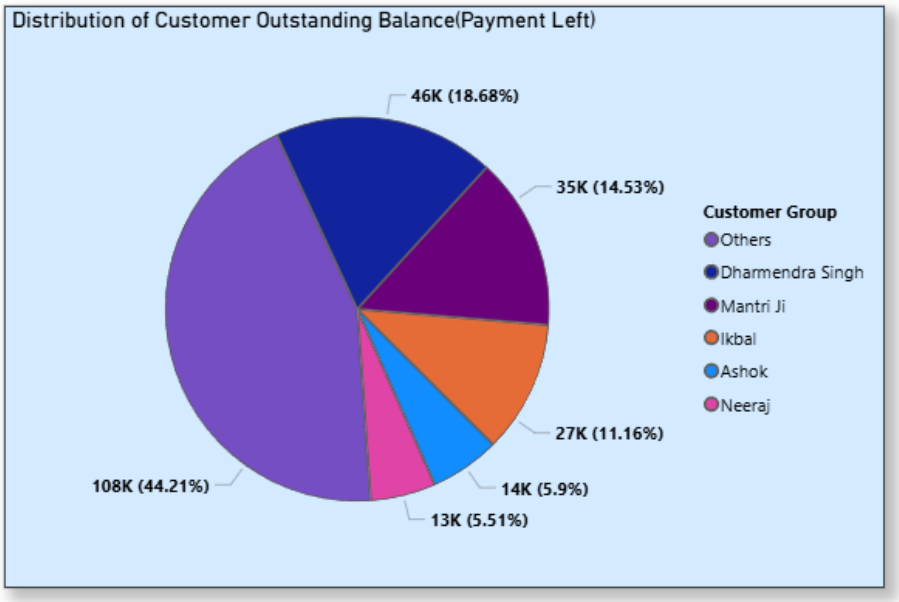


CHART-1 – PIE CHART

- Chart-1 shows that the “Others” category only holds 44.21% of total outstanding credit, reflecting a wide spread across many smaller customers. Among the major contributors, Dharmendra Singh accounts for 18.68 %, followed by Mantri ji (14.53 %),Ikbal by 11.16%, Ashok by 5.9 % and Neeraj(5.51 %).

- Overall, nearly half of the total debt is concentrated among a few key customers, while the rest is distributed across a larger base. This pattern supports the Pareto principle and highlights the importance of focusing on high-priority customers to improve recovery, cash flow, and financial stability.

➢ On the basis of total payments received, the top 10 customers were identified. The same can be seen in the respective chart.

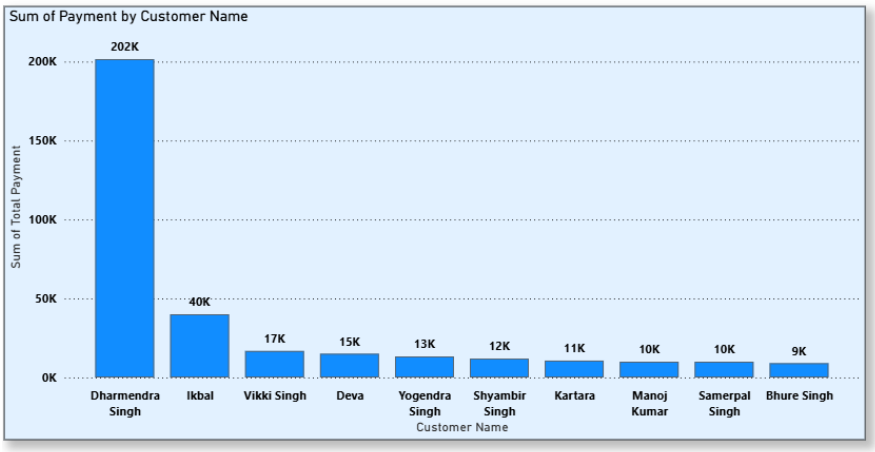


Chart- 2

- Chart-2 shows total payments by customers, clearly indicating that Dharmendra Singh contributed the highest amount, significantly exceeding all others, thus helping identify the most valuable revenue-contributing customer.

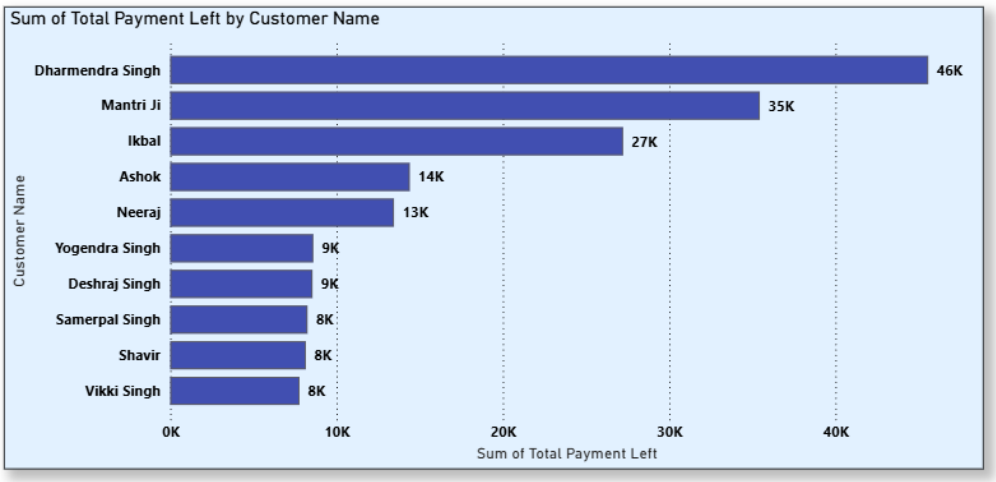


CHART-3

- Chart 3 presents a bar chart of the total outstanding payments by customer, showing a clear descending trend where Dharmendra Singh leads with the highest pending amount (46K), followed by Mantri Ji (35K) and Ikbal by (27K), after which there is a noticeable drop in outstanding values among the remaining customers, with most balances falling below 14K, ,indicating a significant gap between the top defaulters and others and suggesting an uneven distribution of dues that can help prioritize follow-ups based on the magnitude of pending payments.

➢ To understand how the total product price and payments have changed over time, the following chart 'CHART-4' was generated.

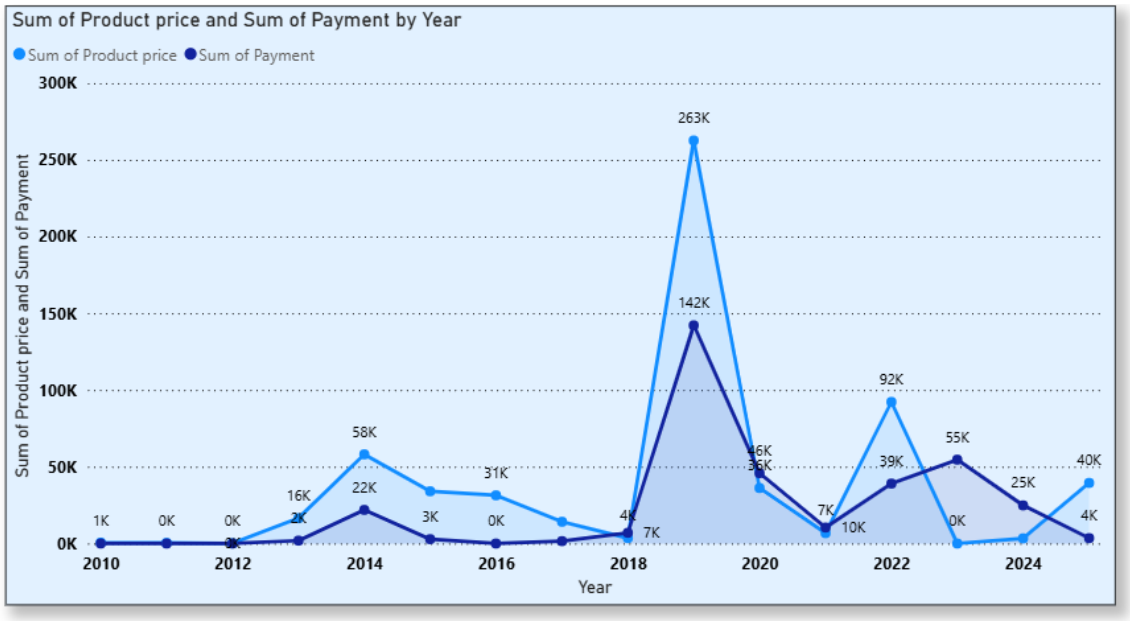


CHART-4–LINECHART

- Line graphs are used to track changes over short and long periods of time.
- The above line chart is created to observe the trend in the total product price and total payment across different years.
- It can be clearly seen that the year 2019 recorded the highest product price (around 263K) as well as the highest payment (around 142K), indicating peak business activity during that period.
- After 2019, there is a noticeable decline in both product price and payment in 2020, followed by fluctuations in the subsequent years.
- Another rise can be observed around 2022, showing a recovery or increase in transactions during that time.
- Overall, the chart highlights that both product price and payment follow a similar trend, suggesting a strong relationship between sales value and actual payments received.

"FREQUENT MRP CHANGE"

➢ Figure 5 is a multiple score card representation of the total valuation shift following the update from Old MRP to New MRP across the production ventory.

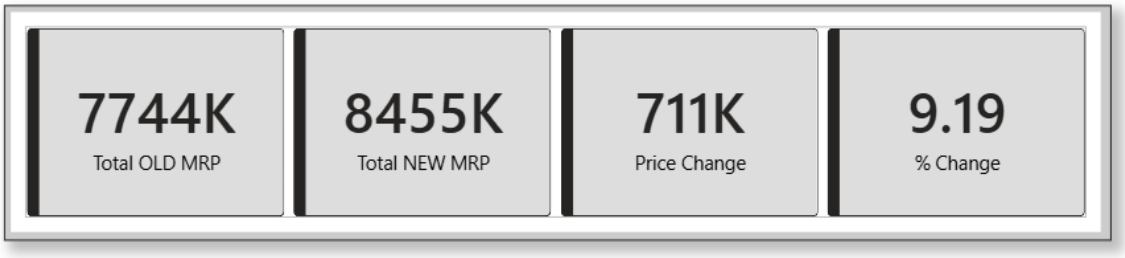


Figure-5

- Scorecard tiles work well in providing a high-level summary of Key Performance Indicators (KPIs). They offer an immediate quantitative snapshot of the financial scale and the percentage of change, allowing for a quick assessment of overall value growth without the complexity of a full trend line.

- From these metrics, it can be seen that the total inventory value rose from a base of ₹ 7744K to 8455K. The absolute difference, represented by the "Sum of Price Increase," shows a total upward valuation of ₹ 711.345K. This confirms a significant inflationary adjustment or a strategic price hike across the board.

- The Percentage Increase MRP stands at 9.19%. This indicates that products saw a percentage increase in their listed price, suggesting a substantial shift in the pricing strategy or a response to rising procurement costs.

- Figure-6 shows the revenue potential and price adjustments categorized by the 'HINDWARE RANGE'. This allows for a granular view of how different product lines contribute to the total valuation, can see whole table by Click here

HINDWARE RANGE	Sum of OLD MRP	Sum of NEW MRP	Price Change	% Change
KITCHEN TAP	20750	25125	4375	21.08
SMALL IDEAL	1980	2390	410	20.71
DRAINS	14250	17110	2860	20.07
FLORA PRIME	67685	81265	13580	20.06
PVD RANGE	15190	18235	3045	20.05
DIVERTER	59300	71035	11735	19.79
DIVERTOR	11680	13890	2210	18.92
CONTESSA NEO	52390	62125	9735	18.58
DIVERTOR BODY	15980	18825	2845	17.80
EDGE	350145	411020	60875	17.39
SHOWER ARM	79430	92410	12980	16.34
ELEMENT	456560	530655	74095	16.23
PRESSMATIC	12040	13990	1950	16.20
ACCESSORIES	99475	115550	16075	16.16
FLUID	71700	82675	10975	15.31
FLORA	90835	104640	13805	15.20
SHOWERS	42980	49450	6470	15.05
DOVE	55900	63855	7955	14.23
HEALTH FAUCET	43995	50155	6160	14.00
AGNESE	78795	89685	10890	13.82
STUDIO	124610	141455	16845	13.52
AVIOR	180240	204125	23885	13.25
CORTO	7980	9000	1020	12.78
OH SHOWER	961855	1084155	122300	12.72
ECLIPSE	8980	10080	1100	12.25
HAND SHOWER	79965	89705	9740	12.18
ARMADA	42300	47365	5065	11.97
ELEMENT NEO	1390	1550	160	11.51
ASPIRO	184055	205220	21165	11.50
Total	**7743690**	**8455035**	**711345**	**9.19**

Figure-6

- The data shows that KITCHEN TAP and SMALL IDEAL had the highest percentage increases (around 21%), while the OH SHOWER range contributed the highest overall value, with a New MRP exceeding 1084K. This suggests pricing was more aggressive in premium ranges, whereas high-volume ranges like OH SHOWER saw a moderate increase of about 12.72%.

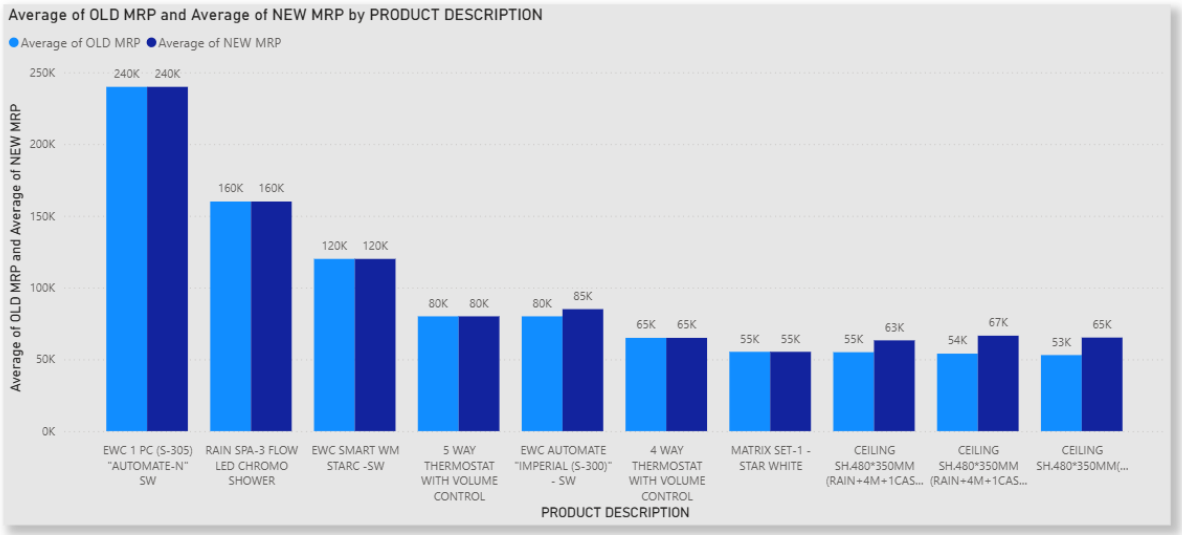


CHART-5

- A clustered bar chart was created to compare the average Old MRP and New MRP by product description, helping visualize pricing trends across individual items. This type of chart clearly shows the gap between old and new prices, especially for top-tier products.

- It is observed that the highest-value item, EWC 1 PC (S-305) "AUTOMATE-N", remained stable at 240K, while products like CEILING SH.480×350MM increased notably from 53K to 65K. This indicates that premium anchor products held steady, whereas mid-to-high range items saw price hikes.

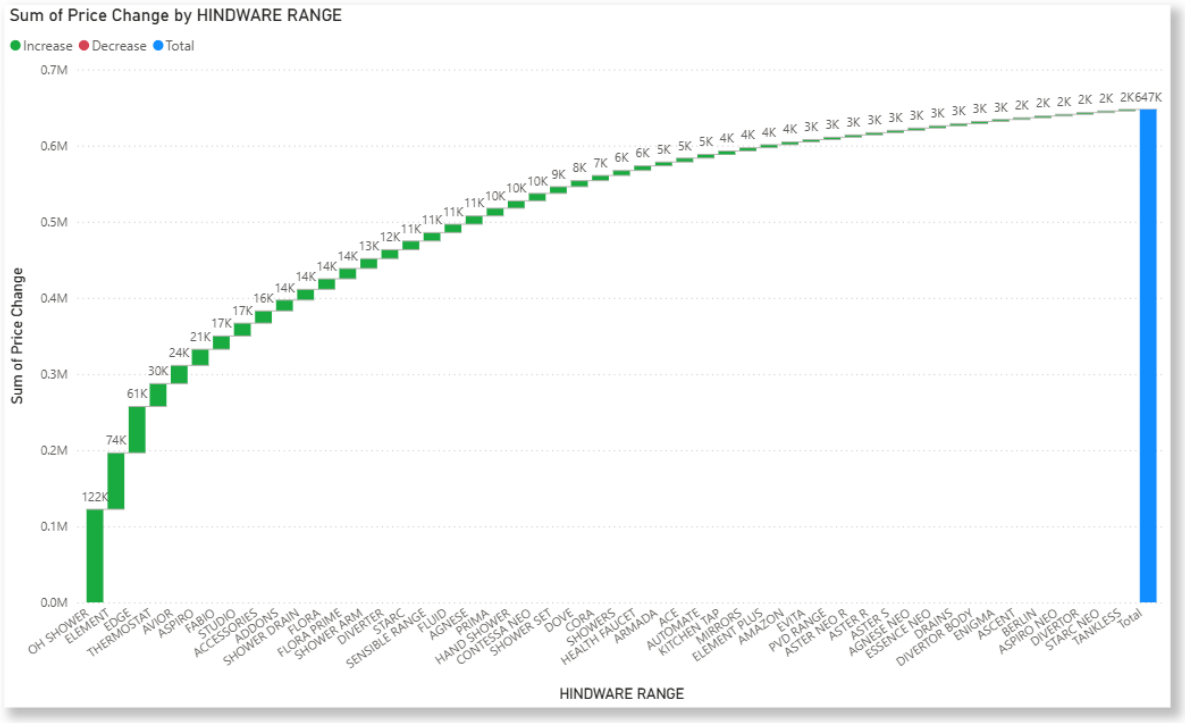


CHART-6

- A Waterfall chart was used to show how the initial value transitions to the final total through a series of intermediate increases, helping visualize the cumulative price change across Hindware ranges. It effectively explains the "bridge" between old and new valuations. To identify which ranges contributed most to the overall ₹647K increase, CHART-6 was created, allowing a clear view of how each category adds to the total.

- The chart shows that the first five categories—OH SHOWER, ELEMENT, EDGE, THERMOSTAT, and AVIOR—are the major contributors, together accounting for over ₹300K of the increase. This indicates that most of the financial impact is concentrated in a few key ranges, highlighted by the steep rise at the beginning. In contrast, smaller contributions from ranges like TANKLESS and DIVERTOR NEO (about ₹2K each) form a long tail, reflecting relatively stable pricing in those categories.

"LOSSES DUE TO BROKEN OR MISSING COMPONENTS"

- Chart-7 is a horizontal bar representation of the financial loss attributed to specific products, likely due to breakage, defects, or "broken parts" within the inventory. Based on the Broken Dataset we have chosen the top 10 Goods sorted in the descending order based on the Loss Value.

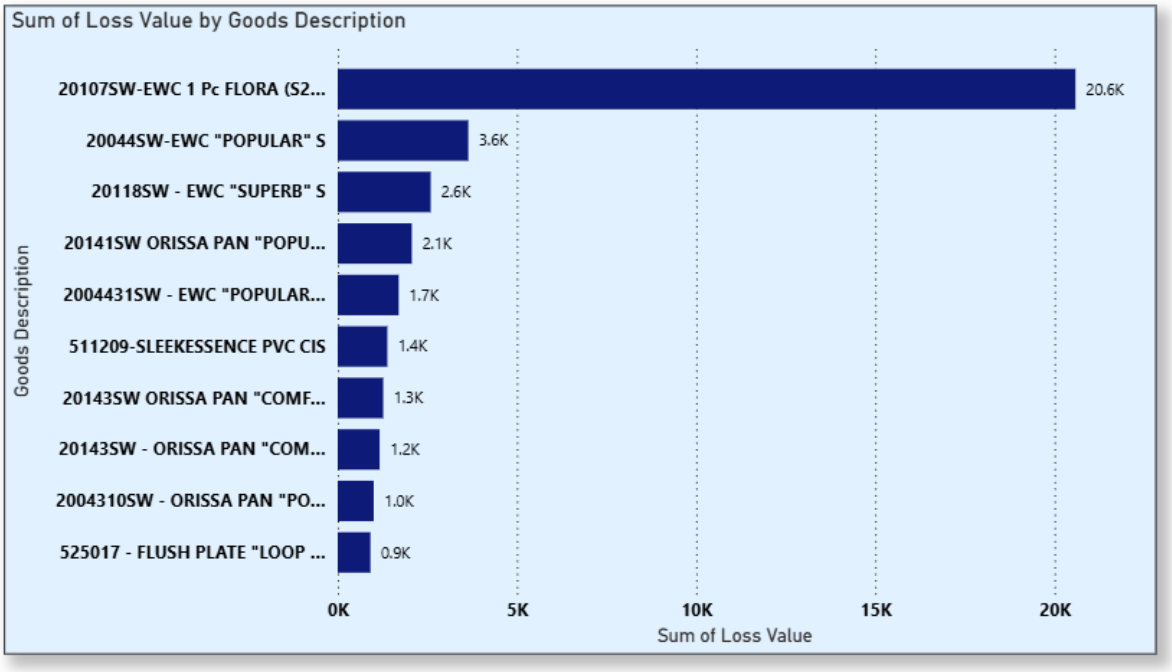


CHART-7

- Horizontal bar charts are useful for ranking items by a specific metric like Loss Value, making it easy to identify high-risk products. The chart shows a highly skewed loss distribution, where a single product, 20107SW-EWC 1 Pc FLO..., contributes about 20.6K in losses—more than Five times the next highest item.

- This insight can help improve stock handling, as EWC units and Orissa Pans dominate the top loss categories, suggesting a need for stricter quality checks and better packaging for these fragile items. While the top 10 products account for significant losses, many items like ORISSA PAN "PO...." and FLUSH PLATE "LOOP...." show comparatively less loss, and by a quick glance on the raw broken dataset shows that most inventory is nearly zero that indicates that inventory is handled well and the issue is concentrated in a small 15–20% of products.

- Chart-8 is a Pareto representation that combines a bar graph (Total Loss) and a line graph (Cumulative Percentage) to identify the most significant factors contributing to inventory loss.

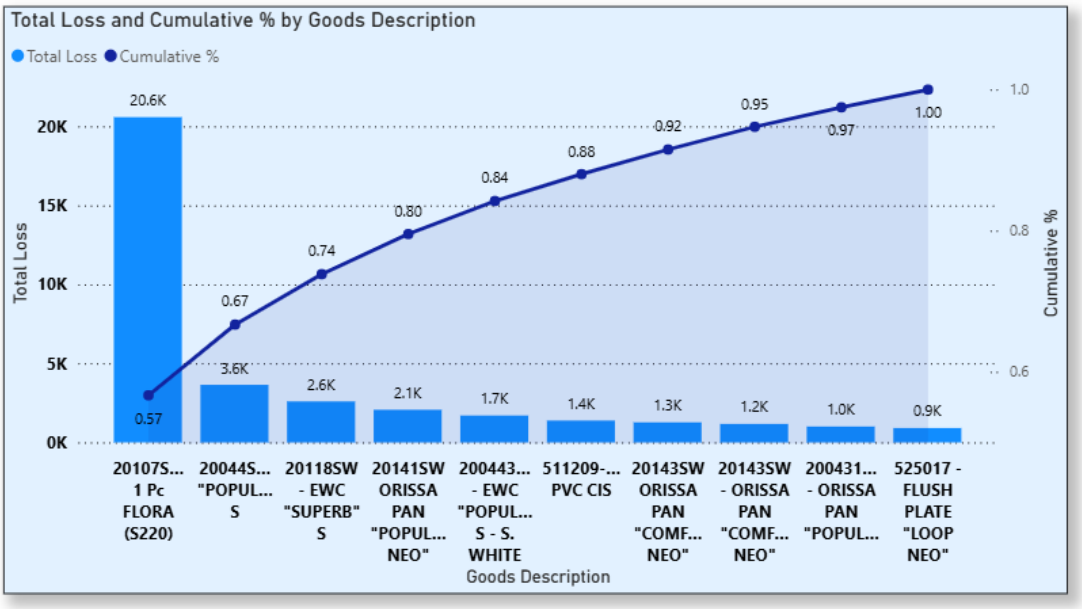


CHART-8

- It is evident that the traditional 80/20 Pareto rule is not perfectly followed in this case, as the cumulative curve reaches around 80% by the 4th–5th item and approaches 100% by the 10th item. This suggests that losses are highly concentrated within a limited number of items rather than being evenly distributed.

- From the chart, it can be seen that a small proportion of products—primarily the first few high-loss items—account for a significant share of the total financial loss. The highest loss is recorded for the 20107SW–EWC 1 Pc FLORA (~20.6K), followed by the next items around 3.6K and 2.6K, with the first 3–4 bars contributing over 70–80% of the total loss.

- In fact, the loss generated by the top item alone (~20.6K) is significantly higher than most other products, many of which individually contribute less than 2K–3K. This clearly indicates that a few high-value items dominate the losses, making them the critical “vital few” that should be prioritized to improve overall cost efficiency and profitability.

- Chart-9 uses a bubble (scatter) plot to compare Damage Rate and Total Loss, with bubble size representing total quantity. It highlights a visible relationship where items with higher damage rates tend to contribute more significantly to financial losses, especially those located in the upper-right region of the chart.

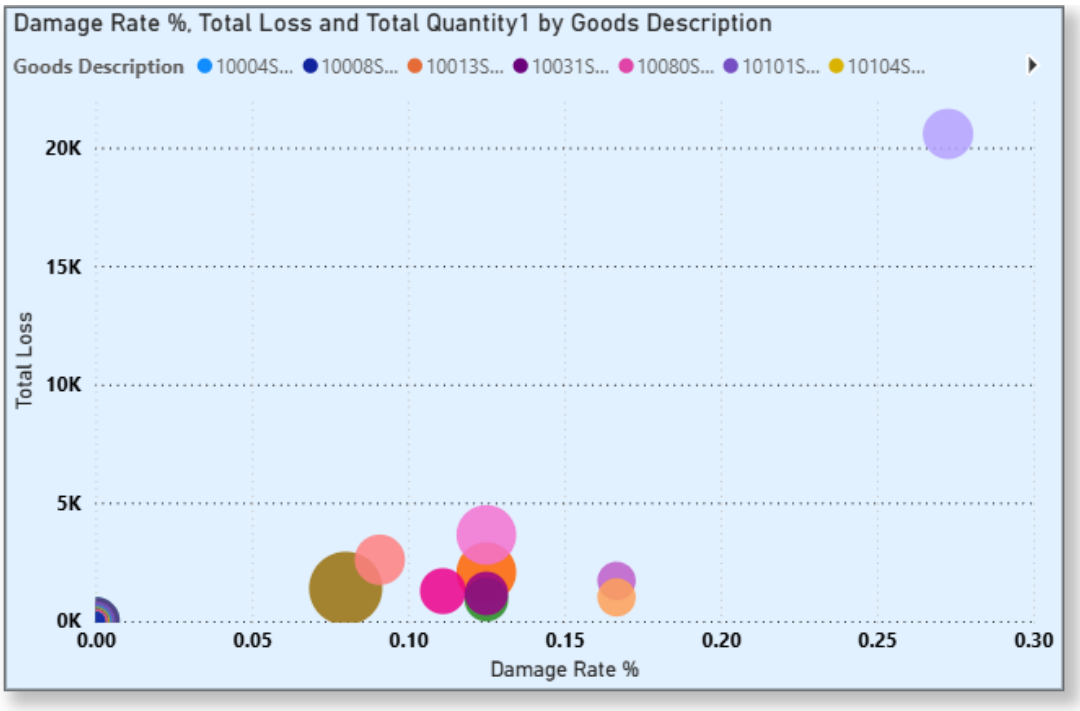


CHART-9

- It can be observed that the highest-loss product shows a loss value of approximately ₹20K–21K, with a damage rate close to 0.28–0.30 (28%–30%), making it a clear outlier. This indicates that this single item contributes disproportionately to total losses compared to others.

- Several other products fall within the mid-range, showing damage rates between 0.10 and 0.15 (10%–15%) and loss values ranging from ₹1K to ₹4K. These items form a cluster, suggesting a moderate but consistent impact on overall losses.

- On the other hand, a few products located near the bottom-left corner have damage rates close to 0%–0.02 (0%–2%) and negligible loss values. This indicates that damage-related issues are not widespread across all products but are concentrated in specific high-risk categories.

- Chart-10 is a Treemap representation of the Sum of Loss Value by Goods Description. Treemaps are useful for visualizing the "part-to-whole" relationship through nested rectangles.

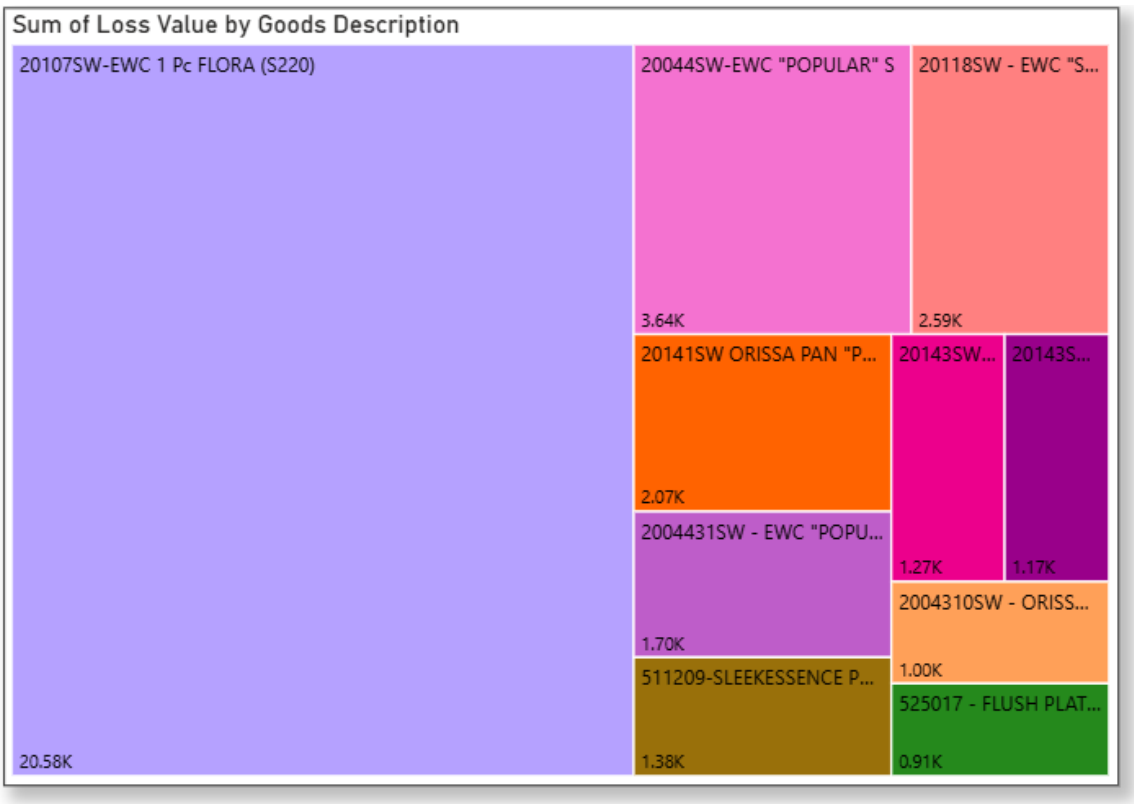


Chart-10

- The treemap clearly shows that the largest block corresponds to 20107SW–EWC 1 Pc FLORA (S220), contributing approximately ₹20.58K to the total loss. This single item dominates the chart, occupying a significantly larger area than all other products combined.

- The next major contributors include items such as 20044SW–EWC “POPULAR” S (~₹3.64K) and 20118SW–EWC (~₹2.59K), followed by others like ORISSA PAN (~₹2.07K) and additional EWC variants ranging between ₹1K–₹1.7K.

- Smaller blocks such as FLUSH PLATE (~₹0.91K) and similar items indicate relatively low financial impact. This highlights that losses are heavily concentrated in a few high-value EWC products, suggesting that improving handling, storage, and quality control for these key items can significantly reduce overall losses.

Interpretation of Results

After reviewing the analysis and findings, the report can now be wrapped up with proposed solutions and recommendations aimed at propelling the business towards progress and benefit. Through the use of diverse graphs and data analysis across various fields, a comprehensive understanding of the data was gained regarding optimizing the financial and operational stability of the retail store.

Following are the interpretations:

1. **Concentrated Debt Risk:** A small group of key customers contributes to over half of the total outstanding debt, while the remaining 44.21% is spread across a wide base of smaller "Others".

2. **Top Revenue Contributor:** Dharmendra Singh was identified as the most valuable customer, significantly exceeding all others in total payments made.

3. **Critical Payment Gaps:** Dharmendra Singh leads with the highest pending amount (46K), highlighting a significant gap between top defaulters and the rest of the customer base.

4. **Historical Peak:** Business activity, in terms of both product price and actual payments received, reached its peak in 2019 before experiencing a decline and subsequent fluctuations.

5. **Inventory Valuation Growth:** The total inventory value saw a substantial upward shift of approximately 711K, representing the MRP increase of 9.19%.

6. **Aggressive Pricing in Premium Ranges:** Price hikes were most aggressive in specialty categories like KITCHEN TAP and SMALL IDEAL (around 21%), whereas high-volume ranges like OH SHOWER saw more moderate increases.

7. **Fragility in Specific Categories:** Financial losses are not distributed evenly; a single category, 20107SW-EWC 1 Pc FLORA, accounts for a massive 20.58K in losses—more than many times any other item.

8. **Correlation Between Damage and Value:** There is a clear relationship observed where products with the highest damage rates (reaching nearly 28%–30%) also contribute to the highest financial losses, indicating that increased damage directly impacts overall loss values.

9. **Pareto of Loss:** Approximately 20% of product types—mainly premium EWC and Pan models—contribute to over 70% of the shop's total financial loss, largely driven by breakage and product defects.

Recommendations

After generalizing all the interpretations, the following recommendations have been made to enhance business performance:

1. **Prioritize Debt Recovery:** Focus collection efforts on the "vital few" high-priority customers like Dharmendra Singh and Mantri Ji to immediately improve cash flow and working capital.

2. **Implement Tiered Credit Limits:** Establish stricter credit limits for the "Others" category to prevent the wide spread of small debts from collectively impacting financial stability.

3. **Sync Selling Prices with MRP Updates:** Ensure that selling prices are updated immediately following revised MRPs to prevent products from being sold at outdated rates, thereby protecting profit margins.

4. **Strategic Focus on High-Volume Ranges:** Since the OH SHOWER range contributes the highest overall inventory value, ensure its stock levels are meticulously managed to avoid stock-outs in this critical volume driver.

5. **Enhance Handling for Fragile Goods:** Implement stricter quality checks and improved packaging/transportation protocols specifically for EWC units and Orissa Pans, as these dominate the loss categories.

6. **Loss Mitigation for Premium Items:** Focus wastage reduction efforts primarily on the FLORA S220 range, as it is the single most "expensive" failure point for the firm.

7. **Monitor Trend Correlations:** Continuously monitor the relationship between sales value and actual payments to ensure the shop does not over-extend credit during periods of high transactional volume.